



// Profile

Technology leader and software engineer/architect with 20+ years delivering complex systems across both the private and public sectors in domains such as climatology, quantum instrumentation, cryptography, blockchain, finance & telecoms. At home setting technical strategy, advising c-suite executives and shaping the processes that let engineering organisations scale. Track record of building a culture of trust where exceptional teams flourish. Member of the Institute of Electrical and Electronics Engineers (IEEE) and the Central Bank Research Association.

// Experience

2021 – Present

Senior Software Consultant

QZabre @ ETH · Zurich, Switzerland

Driving software at an ETH spin-off that builds nanoscale microscopes based on quantum magnetometry. Guided the team's shift to a lightweight, structured agile process; designed and built controllers for photonic and magnetic devices; redesigned the desktop application; and championed the adoption of Rust across the in-house QS3 platform. Advising both the engineering team and the executive suite.

2025 – 2026

Senior Software Consultant

Inversed Tech · Madrid, Spain

A hybrid role across an R&D company specialising in cryptography and distributed systems. As Rust engineer, built the genesis protocol and tooling for a biometric identity platform based on secure multi-party computation. As technical product owner, steered customer-facing projects including a regulated privacy-focused stablecoin and a privacy-preserving loyalty programme. As business development lead, represented the company at industry events, shaped strategy and tracked emerging trends.

2022 – 2024

Head of R&D

Casper Association · Zug, Switzerland

Led R&D in a complex, high-visibility blockchain environment. Partnered with the Head of Engineering to fold research proofs of concept into the core product, and worked with a team of macro-economists to design a blockchain-enabled core banking platform built around an emerging financial standard. Hands-on Rust throughout, applying zero-knowledge cryptography to deliver a ZK rollup and a ZK light client.

2009 – 2022

Software Consultant

Institut Pierre Simon Laplace @ La Sorbonne · Paris, France

Worked at the heart of climate research, extending the IPSL's HPC simulation environment with a real-time, event-driven messaging platform powering automated documentation, live monitoring, performance analytics and machine intelligence. Technical lead of es-doc.org, an international climate-model documentation project, and a driver of process change across the institute.

2019 – 2021

Software Consultant

Casper Labs AG · Zug, Switzerland

Part of the global team that delivered a layer-1 blockchain platform built on DAG-based proof-of-stake consensus. Designed and implemented an event-driven, scalable fuzz-testing platform for surfacing consensus flaws, and streamlined platform integration for external partners. Also led the professional-services team and mentored executives.

2018 – 2019

Software Architect

Trinkler Software · Zug, Switzerland

Helped build a decentralised finance ecosystem for a Zug start-up. Delivered a GraphQL trading API, a feature-rich command-line application, a multi-chain digital wallet and a Vue.js front end, while mentoring junior colleagues and reviewing ecosystem design.

2008 – 2013

Software Architect

NTT Data · Fribourg, Switzerland

Drove technology and process change in the Swiss arm of a global IT services company: introduced agile practice — Scrum, XP, CI/CD — designed a .NET SOA framework and mentored a seven-person development team.

2007 – 2008

Software Architect

British Telecom · Dublin, Ireland

Designed and led development of an interoperable B2B order-processing web service, applying .NET 3.0, domain-driven design and SOA. Also designed a rules-based workflow engine.

2006 – 2007

Software Engineer

DeCare Systems · Cork, Ireland

Designed and led development of a high-volume, real-time enrolment service for a US insurance provider, using an n-tier architecture with WCF communication between tiers.

2006

Software Engineer

SnapOn Diagnostics · Cork, Ireland

Migrated C++ systems to .NET 2.0 under an agile process, and built web services sharing real-time automotive diagnostics across a suite of listening applications. Scrum, XP and TDD throughout.

2005

Software Engineer

Classic Car Collection · Paris, France

Full-stack .NET engineer on a sales and management information system, covering architecture and design, web development, testing and deployment.

1989 – 2001

Earlier Career

Securicor Systems · SSRADU · 3S Electronic Publishing · Customer Interface · Cap Gemini – UK

Software engineering and architecture roles across the UK, spanning n-tier client-server systems, COM frameworks in Visual Basic, database design and technical authorship. Highlights include architecting central London's traffic-violation adjudication system at the head of a twelve-person team (Securicor) and delivering Europe's largest IBM AS-400 project (Cap Gemini).

// Skills

LANGUAGES & FRAMEWORKS

C#PythonRustTypeScriptBashReactVueSvelte

ARCHITECTURE

BlockchainEvent Driven ArchitectureMessage BrokersMicro-ServicesAgentic Systems

AGENTIC (CCA-F CURRICULUM)

Agentic ArchitectureClaude Agent SDKModel Context Protocol (MCP)Prompt EngineeringStructured OutputTool Design & IntegrationContext & Reliability

DATA & MESSAGING

PostgreSQLMongoDBElasticsearchOracleSQL ServerRabbitMQ

DEVOPS & PLATFORMS

DockerKubernetesTerraformAnsibleAWS CloudLinux

PROCESS

AgileScrumXPKanbanTDDCI/CDGitHub

// Education

2026

Claude Certified Architect Foundation (in progress)

Anthropic

2008

Microsoft Certified Software Architect

Microsoft Campus, San Francisco, USA

1989

B.Sc. Economics & Operational Research

University of Leeds, UK

// Other

RESIDENCY

UK citizen with Swiss Residency Permit

LANGUAGES

English (fluent). French (fluent). Italian (advanced). German (average)

TRAVEL

Japan. China. Europe. India. Middle East. North & Central America

INTERESTS

Tennis. Cricket. Rugby. Swimming. Cycling. Singing. Writing. History. Art